

设以二元函数 $z = f(x, y)$ 为例, z 具有所连续偏导数.

$$\text{沿 } x \text{ 轴: } f_x(x, y) = \frac{\partial f(x, y)}{\partial x}$$

$$\text{沿 } y \text{ 轴: } f_y(x, y) = \frac{\partial f(x, y)}{\partial y}$$

单位向量 $\mathbf{l} = (\cos\theta, \sin\theta)$, θ 是与 x 轴正向夹角.

$$\text{沿 } \mathbf{l}: \text{grad} f = f_x(x, y)\cos\theta + f_y(x, y)\sin\theta$$

$$\text{设梯度 } \nabla f = (f_x(x, y), f_y(x, y))$$

$$\begin{aligned} \text{最小/下降最快: } \min(\text{grad} f) &= \min(\nabla f \cdot \mathbf{l}) \\ &= \min(|\nabla f| \cdot |\mathbf{l}| \cdot \cos\alpha) \\ &= \min(|\nabla f| \cdot \cos\alpha), \end{aligned}$$

当 $\alpha = 180^\circ$ 取小值, 是 f 的负方向

$\therefore$ 梯度的负方向是函数下降最快的方向.